Removal



Remove the hot air balloon with red, yellow, and blue panels and white stars, seamlessly inpainting the revealed blue sky.



Replacement



Replace the license plate reading "CAMZ-237" on the blue Hyundai Elantra Sport with a new plate reading "ABC-123".



Color Alteration



Change the color of the pastel-colored backpack with light blue and pink sections, worn by the woman in the yellow sweater.



Portrait Editing



Modify the facial features of the blonde woman wearing a light gray blazer and white collared shirt to exhibit a subtle smile.



Addition



Add a small wooden fishing boat into the video, placing it on the sandy beach where it has just entered the water.





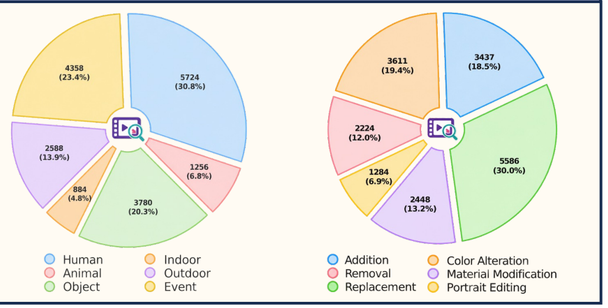
Material Modification



Change the material of the black metal railway signal post in the video into weathered, rusted steel.



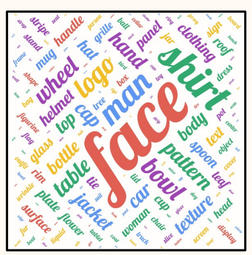
Sample Composition



Word Cloud

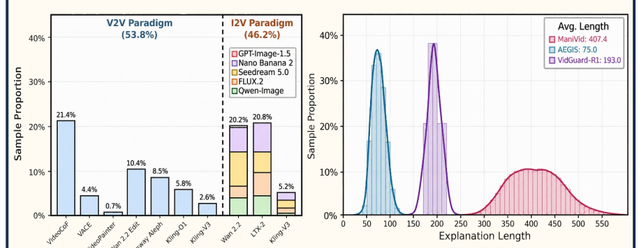


Source Object

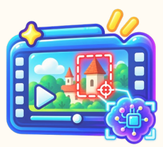


Target Object

Basic Information



Applicable Scenario



<Task 1>
Artifact Grounding



<Task 2>
Anomaly Explanation



<Task 3>
Forgery Detection

Existing Methods Comparison



Add a round, woven rattan side table into the video, placing it on the floor to the right of the sectional sofa's corner.



<Task 1>
Artifact Grounding

<Task 2>
Anomaly Explanation

<Task 3>
Forgery Detection

Image-based Experts

No Temporal Modeling

General MLLMs

Low Level Perception

Video-based Experts

ReStraV: Only Detection VidGuard-R1: No Grounding

Sa2VA: Only Traditional Segmentation Tasks

ManiVidLens (Ours)

- Accurate Classification
- Precise Localization
- Detailed Explanation

Fake

Comprehensive Analysis Report
(1) Boundary Coherence: …
(2) Color and Lighting: …
(3) Texture Discontinuity: …
(4) Temporal Flicker: …

